

The empirical recurrence for OEIS A236078: recovering a conjecture that is not written down

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Abstract

OEIS A236078 records “Empirical recurrence of order 82”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 140$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236078 is “Number of $(n+1) \times (2+1)$ 0..2 arrays with the sum of each 2×2 subblock maximum and minimum lexicographically nondecreasing rowwise and columnwise.”. It has offset 1 and begins

576, 9979, 169680, 2634205, 40097396, 587963980, 8523145663, 121700475056, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:04:15, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 193$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 140$ of the 193, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 140$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 386$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	32	c_{22}	−28120428259271333	c_{43}	25710789246530074803150	c_{64}	229828994
c_2	−151	c_{23}	256503431914537572	c_{44}	−1073606861918888267073	c_{65}	−97111327
c_3	−4918	c_{24}	142271709201868584	c_{45}	−34905940248964229295942	c_{66}	−63801441
c_4	52176	c_{25}	−1714571652619894308	c_{46}	−461563067521075449319	c_{67}	190979487
c_5	243714	c_{26}	−534359559680432919	c_{47}	40766085431105319376372	c_{68}	143159284
c_6	−4921367	c_{27}	9545200643604286050	c_{48}	2978591114435905476289	c_{69}	−2958154
c_7	−1146368	c_{28}	1291521752996483201	c_{49}	−40855983131205581144884	c_{70}	−2567411
c_8	253214693	c_{29}	−44628436501356006626	c_{50}	−5529961410040363907006	c_{71}	35093952
c_9	−418725072	c_{30}	−453976771568876536	c_{51}	35024216209568204602668	c_{72}	36208278
c_{10}	−8503928306	c_{31}	176461319035693533086	c_{52}	6946756192175187517331	c_{73}	−305584
c_{11}	23737045184	c_{32}	−13050525565400939401	c_{53}	−25581372327944834344208	c_{74}	−392127
c_{12}	202727633423	c_{33}	−593498517638064229994	c_{54}	−6676338538000322556945	c_{75}	1816892
c_{13}	−739327337268	c_{34}	73781961370589673639	c_{55}	15845384101194912877410	c_{76}	3144629
c_{14}	−3596806017205	c_{35}	1706076678347780213424	c_{56}	5116636452376260062240	c_{77}	−63520
c_{15}	15868281548742	c_{36}	−249907181191972672232	c_{57}	−8278589228231613347262	c_{78}	−17587
c_{16}	48875756300817	c_{37}	−4207470477929914699672	c_{58}	−3182280960185708081001	c_{79}	78002
c_{17}	−254309095696682	c_{38}	610707721231389792957	c_{59}	3625288416910704261496	c_{80}	61274
c_{18}	−517484361460746	c_{39}	8926117166328829693378	c_{60}	1618266903838624712689	c_{81}	1245
c_{19}	3172957754456562	c_{40}	−1112144927745461163949	c_{61}	−1320725572720788781372	c_{82}	−1002
c_{20}	4304864282237109	c_{41}	−16316287809193298852326	c_{62}	−674256618695977590428		
c_{21}	−31631116951185794	c_{42}	1449846660338408879162	c_{63}	396668143972917906336		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82 + 4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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